

SLIDE 1

Problem & why it matters

Team name: <Team name>

Problem statement (e.g. A2 — Ageing in place): **C1 — FlatFair: HDB resale market intelligence & affordability forecasting**

The problem in one sentence:

Young Singaporeans are making the biggest purchase of their lives in a market where HDB resale prices rose **56% since 2017** while household incomes rose **33%**, with no transparent way to judge whether an asking price, or the repayment behind it, is reasonable.

Who is affected, and how badly (use a number from open data):

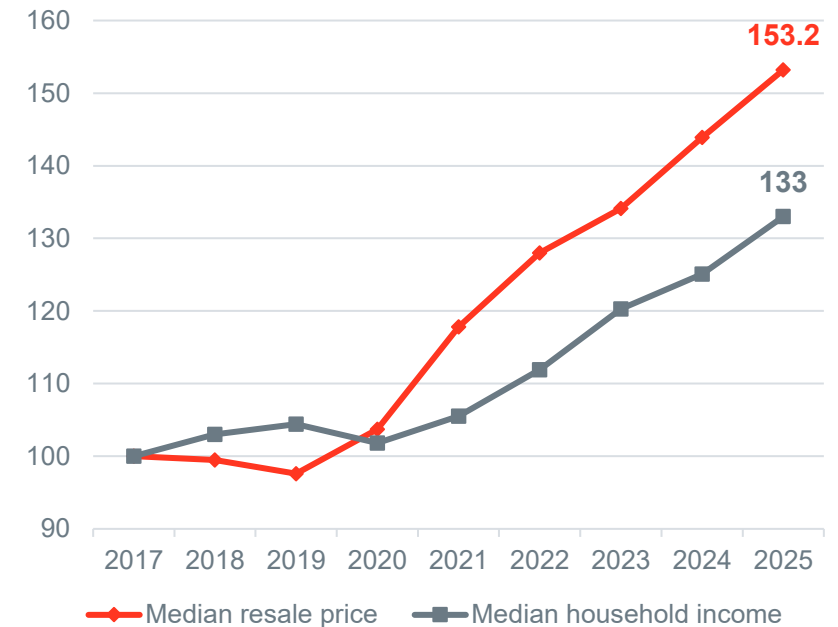
24,662 households bought a resale flat in the last 12 months; **1,847** of those flats sold for \$1 million or more. A median 4-room flat now costs **4.4 years** of the median household's income, up from 3.8 in 2017.

Why it matters now:

After a 50% climb since 2020, prices have flattened (**+1.3% year on year**), so rising values no longer cover an overpayment. Since August 2024 an HDB loan needs a **25% downpayment** instead of 20%, and the October 2024 Standard, Plus and Prime framework has reset expectations again.

Prices have outrun incomes

Index, 2017 = 100



Annual medians. Sources: HDB resale flat prices (data.gov.sg); SingStat M810361 median monthly household employment income.

SLIDE 2

Solution & data

Your solution in one sentence:

FlatFair turns HDB's resale records into a five-step decision tool that takes a buyer from “what is the market doing?” to “is this flat, at this price, sensible for us?”

Who uses it, and what decision it changes:

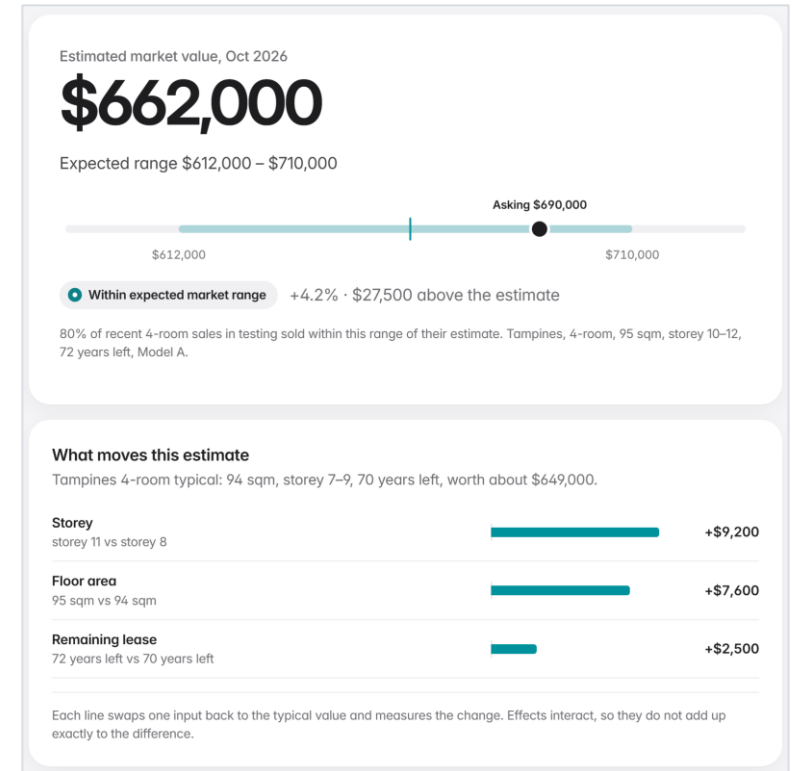
First-time resale buyers, and the counsellors and planners who advise them. It changes **where they look** (towns ranked by what the household can repay), **what they offer** (asking price vs estimated value and the 5 closest sales) and **when they buy** (a tested six-month outlook).

Datasets you will use (name + source, e.g. data.gov.sg):

1. **HDB Resale Flat Prices, Jan 2017 to present** — data.gov.sg d_8b84c4ee...6abc, 241,822 sales
2. **Household Employment Income, key indicators (M810361)** — SingStat Table Builder
3. **HDB Annual Report 2025 · Population by Planning Area** — HDB, SingStat (supply and demand context)

What makes your approach different from a generic dashboard:

It values a specific flat, not just the market: an 80% price range tested on **13,589 sales it never saw** (median error **3.9%** vs 9.5% for the \$/sqm rule of thumb), plus repayment against the 30% MSR cap. Every model must beat a simple baseline: when a 3-month average out-forecast our ML model, we published the average.

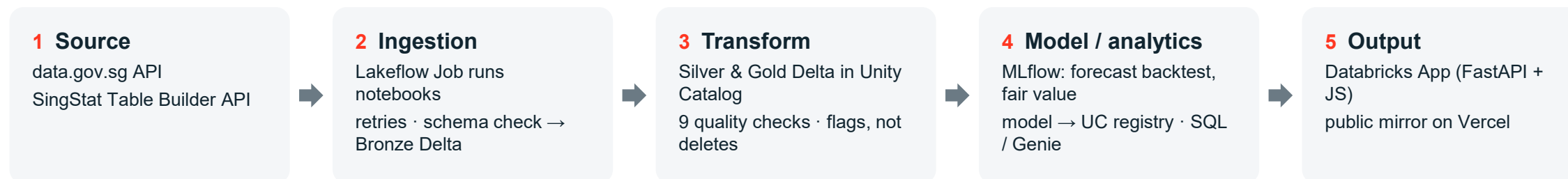


Working prototype: a Tampines 4-room asking \$690,000 sits within the expected \$612k–\$710k range. flatfair-nine.vercel.app

SLIDE 3

Databricks architecture & impact

Pipeline: source → ingestion → transform → model / analytics → output



(Name the Databricks components) **Lakeflow Jobs · Delta Lake · Unity Catalog (lineage, tags, model registry) · MLflow · Databricks SQL / AI-BI · Genie · Databricks Apps**

The user-facing output you will demo:

The FlatFair app: market explorer, six-month forecast with its range, affordability with “Where can I afford?”, fair value with comparable sales, and town comparison.
Already working end to end.

Measurable impact if it worked:

On a typical \$630k flat the pricing blind spot narrows from **~\$60k to ~\$25k** (rule-of-thumb vs model median error). Every buyer sees repayment against the 30% MSR cap before committing, refreshed monthly from open data.

What you can realistically finish in the two-week sprint:

Pipeline, both models, the app and 66 tests are built. Sprint: run it as a Lakeflow Job in the final workspace, deploy on Databricks Apps, add a Genie space and dashboard, test with five first-time buyers.